

5.1 Multiple Qubits

Saturday, November 1, 2025 12:38 PM

Let $|\psi_1\rangle = \alpha_1|0\rangle + \beta_1|1\rangle$ and $|\psi_2\rangle = \alpha_2|0\rangle + \beta_2|1\rangle$, find $|\psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle$

$$\begin{aligned}
 |\psi\rangle &= \alpha_1\alpha_2|00\rangle + \alpha_1\beta_2|01\rangle + \beta_1\alpha_2|10\rangle + \beta_1\beta_2|11\rangle \\
 &= c_0|0\rangle + c_1|1\rangle + c_2|2\rangle + c_3|3\rangle \\
 &= \sum_{i=0}^3 c_i|i\rangle = \sum_{i=0}^{N-1} c_i|i\rangle \quad N=2^2=4
 \end{aligned}$$

2-qubit

$$\begin{aligned}
 n\text{-qubit} \Rightarrow N=2^n-1 \quad |\psi\rangle &= \sum_{i=0}^{N-1} c_i|i\rangle \quad \langle\psi| = \sum_{i=0}^{N-1} \bar{c}_i\langle i|
 \end{aligned}$$

e.g. 3-qubit $\Rightarrow N=2^3-1=7 \Rightarrow |\psi\rangle = \sum_{i=0}^7 c_i|i\rangle$

$$\begin{aligned}
 &= c_0|0\rangle + c_1|1\rangle + c_2|2\rangle + \dots + c_7|7\rangle \\
 &= c_1|000\rangle + c_1|001\rangle + c_2|010\rangle + \dots + c_7|111\rangle
 \end{aligned}$$

Power notation

$$|00\dots 0\rangle = \underbrace{|0\rangle \otimes |0\rangle \otimes \dots \otimes |0\rangle}_{n\text{-times}} = |0\rangle^{\otimes n}$$

Inner Product of standard Bases

$|i\rangle$ are $|j\rangle$ are bases

$$\langle i | j \rangle = \begin{cases} 1 & \text{if } i=j \\ 0 & \text{if } i \neq j \end{cases}$$

Example 5.1.1: Compute $\langle -+ | 10 \rangle$

Solution:

$$\begin{aligned}
 \langle -+ | 10 \rangle &= \frac{1}{2} (\langle 00 | + \langle 01 | - \langle 10 | - \langle 11 |) | 10 \rangle \\
 \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} &= \frac{1}{2} (\underbrace{\langle 00 | 10 \rangle}_0 + \underbrace{\langle 01 | 10 \rangle}_0 - \underbrace{\langle 10 | 10 \rangle}_{(-1)} - \underbrace{\langle 11 | 10 \rangle}_0) \\
 &= -\frac{1}{2}
 \end{aligned}$$

Partial Measurement

Example 5.1.2: Consider the following quantum state

$$|\psi\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{2}|01\rangle + \frac{\sqrt{3}}{4}|10\rangle + \frac{1}{4}|11\rangle$$

- Measure the left qubit in the state $|0\rangle$. What quantum state does the system collapse to after the measurement?
- Measure the right qubit in the state $|1\rangle$. What quantum state does the system collapse to after the measurement?

① $P_{0-} = \left|\frac{1}{\sqrt{2}}\right|^2 + \left|\frac{1}{2}\right|^2 = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$

Collapse $\Rightarrow |\psi\rangle = \frac{\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{2}|01\rangle}{\sqrt{3/4}} = \sqrt{\frac{2}{3}}|00\rangle + \frac{1}{\sqrt{3}}|01\rangle$

② $P_{1+} = \left|\frac{1}{2}\right|^2 + \left|\frac{1}{4}\right|^2 = \frac{1}{4} + \frac{1}{16} = \frac{5}{16}$

Collapse $\Rightarrow |\psi\rangle = \frac{\frac{1}{2}|01\rangle + \frac{1}{4}|11\rangle}{\sqrt{5/4}} = \frac{2}{\sqrt{5}}|01\rangle + \frac{1}{\sqrt{5}}|11\rangle$

Example 5.1.3: Consider the following quantum state

$$|\psi\rangle = \frac{1+i}{\sqrt{6}}|000\rangle + \frac{i}{2}|010\rangle + \frac{1-i}{\sqrt{12}}|011\rangle + \frac{1-i}{2\sqrt{2}}|111\rangle$$

Measure the left and middle qubits. Find all possible out comes.

$$\frac{2}{2} = 1$$

$P_{00} = \left|\frac{1+i}{\sqrt{6}}\right|^2 = \frac{2}{6} = \frac{1}{3} \Rightarrow \frac{1+i}{\sqrt{2}}|00\rangle$

$P_{01} = \left|\frac{i}{2}\right|^2 + \left|\frac{1-i}{\sqrt{12}}\right|^2 = \frac{1}{4} + \frac{2}{12} = \frac{5}{12} \Rightarrow \sqrt{\frac{3}{5}} i |010\rangle + \frac{1-i}{\sqrt{5}} |011\rangle$

$P_{10} = 0$

$P_{11} = \left|\frac{1-i}{2\sqrt{2}}\right|^2 = \frac{2}{8} = \frac{1}{4} \Rightarrow \frac{1-i}{\sqrt{2}} |111\rangle$

$P_{00} + P_{01} + P_{10} + P_{11} = \frac{1}{3} + \frac{5}{12} + \frac{1}{4} = \frac{12}{12} = 1$

Entanglement

Definition 5.1.1: A quantum state $|\psi\rangle$ of a composite system (composed of two or more subsystems) is called **entangled** if it cannot be expressed as a product of states of its subsystems. Formally, if we have two subsystems described by states $|\psi_1\rangle$ and $|\psi_2\rangle$, the composite state can be expressed as a tensor product: $|\psi\rangle \neq |\psi_1\rangle \otimes |\psi_2\rangle$

Example 5.1.4: Consider the quantum state

a. $|\psi\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$.

b. $|\phi\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) = \frac{1}{\sqrt{2}}(|0\rangle|1\rangle + |1\rangle|0\rangle)$

Ⓐ $\frac{1}{2}(|0\rangle(|0\rangle + |1\rangle) + |1\rangle(|0\rangle + |1\rangle))$

$= \frac{1}{2}(|0\rangle + |1\rangle)(|0\rangle + |1\rangle)$ not entangled
 $|\psi_1\rangle \quad \otimes \quad |\psi_2\rangle$

$= \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle \otimes |+\rangle$

Result: If $|\psi_1\rangle = \alpha_1|0\rangle + \beta_1|1\rangle$ and $|\psi_2\rangle = \alpha_2|0\rangle + \beta_2|1\rangle$, and $|\psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle$

$$|\psi\rangle = c_0|00\rangle + c_1|01\rangle + c_2|10\rangle + c_3|11\rangle$$

This state is **separable** if and only if

$100 \rightarrow c_0c_3 = c_1c_2$
 $\Leftrightarrow \frac{0}{\sqrt{2}} \cdot \frac{0}{\sqrt{2}} = \frac{0}{\sqrt{2}} \cdot \frac{0}{\sqrt{2}} \Rightarrow 0 = \frac{1}{2}$ False

The **Bell's** states or Einstein, Podolsky, and Rosen (**EPR**) pairs entanglement states

$$|B_{00}\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

$$|B_{01}\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$$

$$|B_{10}\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$

$$|B_{11}\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

5.2 Multiple Qubit Gates

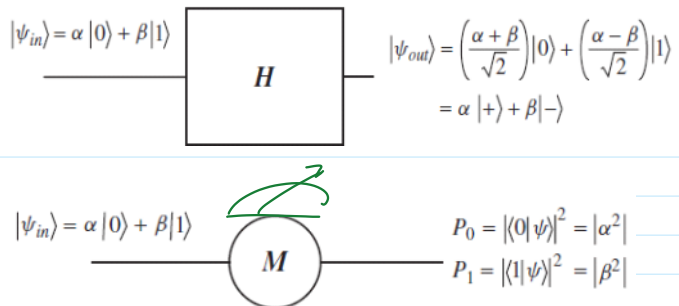
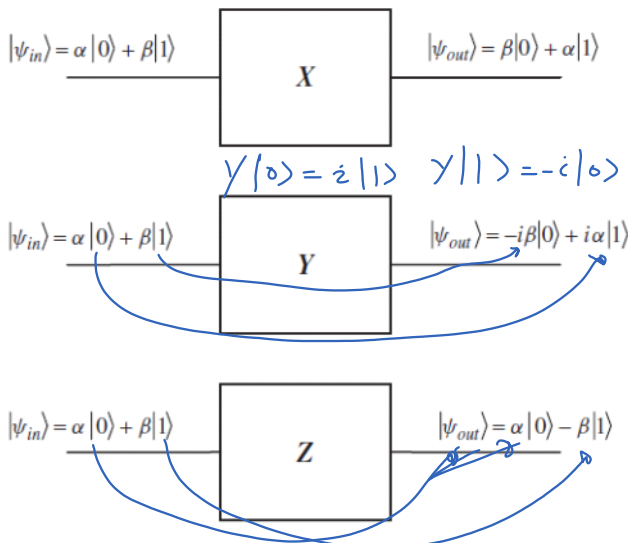
Saturday, November 1, 2025 12:44 PM

Single qubit gate

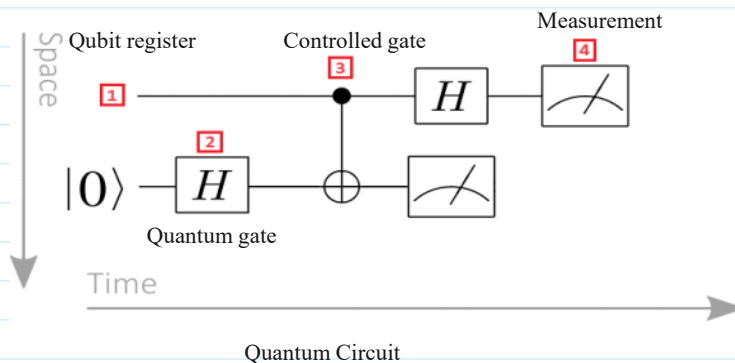
$$X|\psi_{in}\rangle = \alpha X|0\rangle + \beta X|1\rangle$$

$$H|\psi_{in}\rangle = \alpha \left(\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \right) + \beta \left(\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \right)$$

$$= \frac{1}{\sqrt{2}}(\alpha + \beta)|0\rangle + \frac{1}{\sqrt{2}}(\alpha - \beta)|1\rangle$$



Note: The number of inputs of a quantum gate are equal to the number of outputs of a quantum gate because the quantum operations are unitary and hence reversible.



Applying single qubit gate on multi-qubit state

Example 5.2.1: Find the output of the circuit $(H \otimes I)(|01\rangle)$

$$|0\rangle \xrightarrow{H} \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

$$|1\rangle \xrightarrow{I} |1\rangle$$

$$(H \otimes I)(|0\rangle \otimes |1\rangle)$$

$$= H|0\rangle \otimes I|1\rangle$$

$$= \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |1\rangle$$

$$= \frac{1}{\sqrt{2}}(|01\rangle + |11\rangle)$$

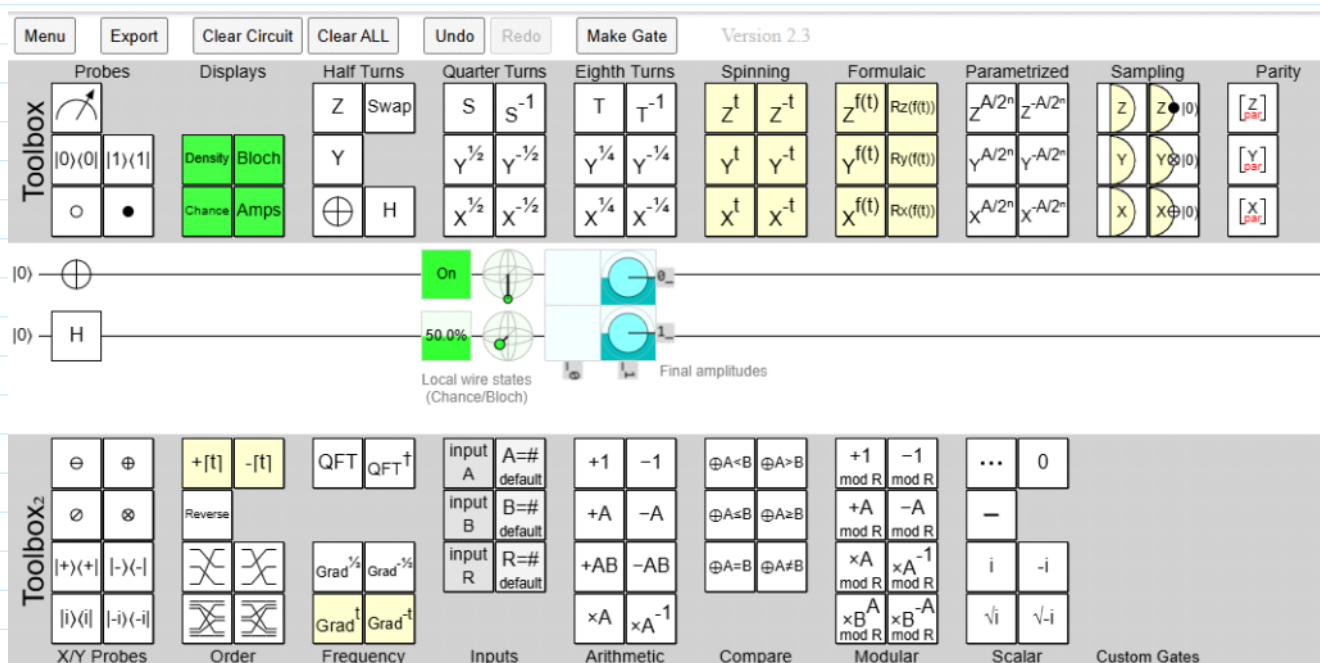
Example 5.2.2: Consider $|\psi\rangle = \frac{1}{4}|00\rangle + \frac{1}{2}|01\rangle + \frac{1}{\sqrt{2}}|10\rangle + \frac{\sqrt{3}}{4}|11\rangle$, find the output of the

Example 5.2.2: Consider $|\psi\rangle = \frac{1}{4}|00\rangle + \frac{1}{2}|01\rangle + \frac{1}{\sqrt{2}}|10\rangle + \frac{\sqrt{3}}{4}|11\rangle$, find the output of the circuit $(H \otimes X)(|\psi\rangle)$

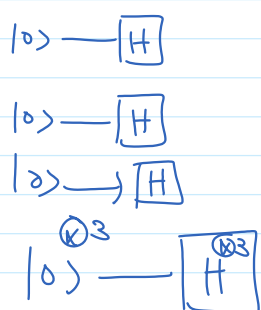
$$\begin{aligned}
 (H \otimes X)|\psi\rangle &= \frac{1}{4} \left(\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \right) |1\rangle + \frac{1}{2} \left(\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \right) |0\rangle + \frac{1}{\sqrt{2}} \left(\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \right) |1\rangle + \frac{\sqrt{3}}{4} \left(\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \right) |0\rangle \\
 &= \frac{1}{4\sqrt{2}} (|01\rangle + |11\rangle) + \frac{1}{2\sqrt{2}} (|00\rangle + |10\rangle) + \frac{1}{2} (|01\rangle - |11\rangle) + \frac{\sqrt{3}}{4\sqrt{2}} (|00\rangle - |10\rangle) \\
 &= \frac{1}{4\sqrt{2}} + \frac{1}{2\sqrt{2}} = \frac{2 + \sqrt{3}}{4\sqrt{2}} |00\rangle + \frac{1 + 2\sqrt{2}}{4\sqrt{2}} |01\rangle + \frac{2 - \sqrt{3}}{4\sqrt{2}} |10\rangle + \frac{1 - 2\sqrt{2}}{4\sqrt{2}} |11\rangle
 \end{aligned}$$

Example 5.2.3: Quirk for the circuit $(H \otimes X)(|00\rangle)$

[https://algassert.com/quirk#circuit={%22cols%22:\[\[%22X%22,%22H%22\]\]}](https://algassert.com/quirk#circuit={%22cols%22:[[%22X%22,%22H%22]]})



Example 5.2.4: Compute $H^{\otimes 3}|0\rangle^{\otimes 3}$



$$\begin{aligned}
 H^{\otimes 3}|0\rangle^{\otimes 3} &= \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \otimes \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \otimes \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \\
 &= \frac{1}{\sqrt{2^3}} (|000\rangle + |001\rangle + |010\rangle + |011\rangle + |100\rangle + |101\rangle + |110\rangle + |111\rangle)
 \end{aligned}$$

$$\begin{aligned}
 (|0\rangle + |1\rangle)(|0\rangle + |1\rangle) &= |00\rangle + |01\rangle + |10\rangle + |11\rangle
 \end{aligned}$$

$$111 \dots 111 = \frac{1}{\sqrt{2^3}} (|0\rangle + |1\rangle + |2\rangle + |3\rangle + |4\rangle + |5\rangle + |6\rangle + |7\rangle)$$

$$H|x\rangle = \frac{1}{\sqrt{2}}(|0\rangle + (-1)^x |1\rangle) = \frac{1}{\sqrt{2^n}}(|0\rangle + |1\rangle + |2\rangle + |3\rangle + |4\rangle + |5\rangle + |6\rangle + |7\rangle)$$

x is -1 qubit

$$= \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle$$

$2^n = N$

$$H^{\otimes n} |0\rangle^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle$$

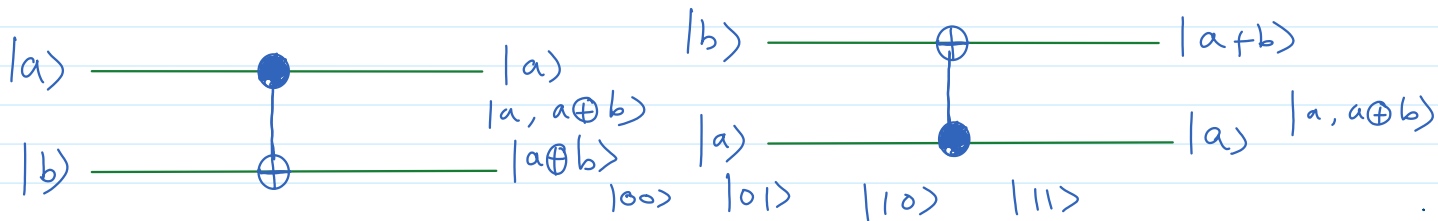
Two-Qubit Quantum Gates:

The **controlled-NOT** gate or the **CNOT** gate: it inverts the right qubit if the left qubit is 1. The left qubit is called the control qubit and the right is called the target.

$$\begin{aligned} CNOT|00\rangle &= |00\rangle \\ CNOT|01\rangle &= |01\rangle \\ CNOT|10\rangle &= |11\rangle \\ CNOT|11\rangle &= |10\rangle \end{aligned}$$

$CNOT|ab\rangle = |a, a \oplus b\rangle$

Controlled
target



Matrix representation of CNOT gate

$$CNOT = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Example 5.2.5: Describe a circuit that will generate the Bell states.

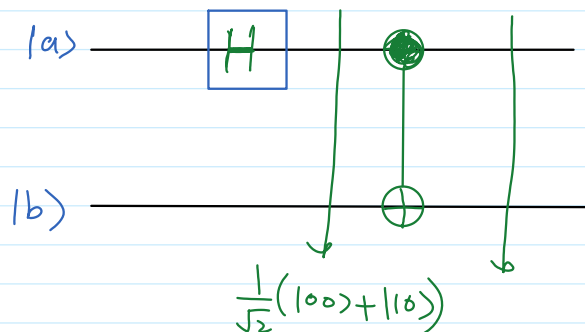
$$|B_{00}\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

$$|B_{10}\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$$

$$|B_{01}\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$

$$|B_{11}\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

B_{ab}



$$B_{00} = H|0\rangle|0\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$$

$$CNOT\left(\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)\right) = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

$$CNOT(H \otimes I |ab\rangle) = H|0\rangle|0\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

Example 5.2.6: Apply Hadamard gate to both sides of CNOT gate, i.e. $(H \otimes H)CNOT(H \otimes H)$. Construct the matrix representation of this circuit

~ab

Example 5.2.6: Apply Hadamard gate to both sides of CNOT gate, i.e. $(H \otimes H)CNOT(H \otimes H)$. $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \otimes \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Construct the matrix representation of this circuit

$$CNOT = \begin{matrix} & |00\rangle & |01\rangle & |11\rangle & |10\rangle \\ \begin{matrix} |00\rangle \\ |01\rangle \\ |11\rangle \\ |10\rangle \end{matrix} & \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \end{matrix}$$

$$\frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}$$

Example 5.2.7: Prove the identity $CNOT(X \otimes I) = (X \otimes X)CNOT$

$$|\psi\rangle = \alpha_1 |00\rangle + \alpha_2 |01\rangle + \alpha_3 |10\rangle + \alpha_4 |11\rangle$$

$$\begin{aligned} CNOT(X \otimes I) &= CNOT(\alpha_1 |10\rangle + \alpha_2 |11\rangle + \alpha_3 |00\rangle + \alpha_4 |01\rangle) \\ &= \alpha_1 |11\rangle + \alpha_2 |10\rangle + \alpha_3 |00\rangle + \alpha_4 |01\rangle \end{aligned}$$

$$\begin{aligned} (X \otimes X)CNOT|\psi\rangle &= (X \otimes X)(\alpha_1 |10\rangle + \alpha_2 |01\rangle + \alpha_3 |11\rangle + \alpha_4 |10\rangle) \\ &= \alpha_1 |11\rangle + \alpha_2 |10\rangle + \alpha_3 |00\rangle + \alpha_4 |01\rangle \end{aligned}$$

5.3 Gate Decomposition

Saturday, November 8, 2025 9:05 PM

The **controlled- U** gate: applies some quantum gate U to the right qubit if the left qubit is 1.

$$\begin{aligned} CU|00\rangle &= |00\rangle, \\ CU|01\rangle &= |01\rangle, \\ CU|10\rangle &= |1\rangle \otimes U|0\rangle, \\ CU|11\rangle &= |1\rangle \otimes U|1\rangle. \end{aligned}$$

$$R_k = \begin{bmatrix} 1 & 0 \\ 0 & e^{ik} \end{bmatrix}$$

Example 5.3.1: Consider $U|0\rangle = a|0\rangle + b|1\rangle$, $U|1\rangle = c|0\rangle + d|1\rangle$, compute $CU|00\rangle$, $CU|01\rangle$, $CU|10\rangle$, and $CU|11\rangle$

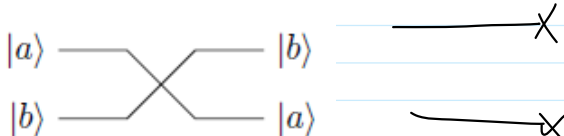
$$\begin{aligned} CU|00\rangle &= |00\rangle \\ CU|01\rangle &= |01\rangle \\ CU|10\rangle &= |1\rangle U|0\rangle = |1\rangle (a|0\rangle + b|1\rangle) \\ &= a|10\rangle + b|11\rangle \\ CU|11\rangle &= |1\rangle U|1\rangle = |1\rangle (c|0\rangle + d|1\rangle) \\ &= c|10\rangle + d|11\rangle \end{aligned}$$

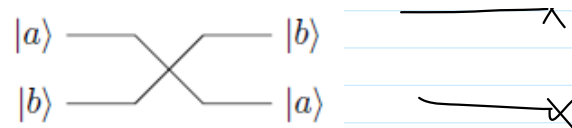
Example 5.3.2: Find the controlled- U gates if $U = X$, and $U = Z$

$$\begin{aligned} U = X \quad & \begin{aligned} C-X|00\rangle &= |00\rangle \\ C-X|01\rangle &= |01\rangle \\ C-X|10\rangle &= |1\rangle X|0\rangle = |11\rangle \\ C-X|11\rangle &= |1\rangle X|1\rangle = |10\rangle \end{aligned} \quad C-X \equiv CNOT \end{aligned}$$

$$\begin{aligned} U = Z: \quad & \begin{aligned} C-Z|00\rangle &= |00\rangle \\ C-Z|01\rangle &= |01\rangle \\ C-Z|10\rangle &= |1\rangle Z|0\rangle = |10\rangle \\ C-Z|11\rangle &= |1\rangle Z|1\rangle = -|11\rangle \end{aligned} \end{aligned}$$

The **SWAP** gate: is simply swaps the two qubits:

$$SWAP|ab\rangle = |ba\rangle$$


$$SWAP|ab\rangle = |ba\rangle$$


Example 5.3.3: Construct the matrix representation of the *SWAP* gate

$$\begin{aligned} SWAP|00\rangle &= |00\rangle \\ SWAP|01\rangle &= |10\rangle \\ SWAP|10\rangle &= |01\rangle \\ SWAP|11\rangle &= |11\rangle \end{aligned} \Rightarrow M = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Note 5.3.4: Show that the *SWAP* gate can be represented of *CNOT*₁₀=*CNOT* and *CNOT*₀₁ gate

$$SWAP = (CNOT)(CNOT_{01})(CNOT)$$

$$\begin{aligned} |\psi_0\rangle &= \alpha_1|00\rangle + \alpha_2|01\rangle + \alpha_3|10\rangle + \alpha_4|11\rangle \\ \textcircled{1} \quad |\psi_1\rangle &= CNOT|\psi_0\rangle = \alpha_1|00\rangle + \alpha_2|01\rangle + \alpha_3|11\rangle + \alpha_4|10\rangle \\ \textcircled{2} \quad |\psi_2\rangle &= CNOT_{01}|\psi_1\rangle = \alpha_1|00\rangle + \alpha_2|11\rangle + \alpha_3|01\rangle + \alpha_4|10\rangle \\ \textcircled{3} \quad |\psi_3\rangle &= CNOT|\psi_2\rangle = \alpha_1|00\rangle + \alpha_2|10\rangle + \alpha_3|01\rangle + \alpha_4|11\rangle \\ \hline SWAP|\psi_0\rangle &= \alpha_1|00\rangle + \alpha_2|10\rangle + \alpha_3|01\rangle + \alpha_4|11\rangle \end{aligned}$$

$$\begin{aligned} CNOT_{01}|00\rangle &= |00\rangle \\ CNOT_{01}|01\rangle &= |11\rangle \\ CNOT_{01}|10\rangle &= |10\rangle \\ CNOT_{01}|11\rangle &= |01\rangle \end{aligned}$$

$$\begin{aligned} SWAP|ab\rangle &= |ba\rangle \\ CNOT(CNOT_{01})CNOT|ab\rangle & \\ CNOT(CNOT_{01})|a\rangle|a\oplus b\rangle & \\ CNOT(|a\oplus a\oplus b\rangle|a\oplus b\rangle) & \\ = CNOT|b\rangle|a\oplus b\rangle & \\ = |b\rangle|a\oplus b\oplus b\rangle = |ba\rangle = SWAP|ab\rangle & \end{aligned}$$

$$\begin{array}{c|c} a \oplus a = 0 & 0 + b = b \\ \hline a & b \\ 0 & 0 \\ 1 & 1 \end{array}$$

Example 5.3.4: Entanglion contains four yellow planets besides the Bell States.

$$\begin{aligned} |\omega_0\rangle &= \frac{1}{2}(|\underline{00}\rangle - |\underline{01}\rangle + |\underline{10}\rangle + |\underline{11}\rangle), & |\omega_0\rangle &= \frac{1}{2}(\underline{|\underline{00}\rangle}(\underline{|\underline{00}\rangle} - \underline{|\underline{01}\rangle}) + \underline{|\underline{10}\rangle}(\underline{|\underline{00}\rangle} + \underline{|\underline{01}\rangle})) \\ |\omega_1\rangle &= \frac{1}{2}(-|\underline{00}\rangle + |\underline{01}\rangle + |\underline{10}\rangle + |\underline{11}\rangle), \\ |\omega_2\rangle &= \frac{1}{2}(|\underline{00}\rangle + |\underline{01}\rangle + |\underline{10}\rangle - |\underline{11}\rangle), \\ |\omega_3\rangle &= \frac{1}{2}(|\underline{00}\rangle + |\underline{01}\rangle - |\underline{10}\rangle + |\underline{11}\rangle). \end{aligned}$$

- Apply the *SWAP* gate to $|\omega_0\rangle$, what is the result state?
- Apply the *X* to the left qubit of $|\omega_1\rangle$

a) Apply the *SWAP* gate to $|\omega_0\rangle$, what is the result state?

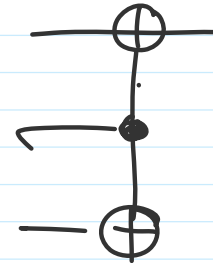
b) Apply the *X* to the left qubit of $|\omega_1\rangle$

$$\text{SWAP}|\omega_0\rangle = \frac{1}{2}(|00\rangle - |10\rangle + |01\rangle + |11\rangle) = \omega_3$$

$$X \otimes I |\omega_1\rangle = \frac{1}{2}(-|10\rangle + |11\rangle + |00\rangle + |01\rangle) = \omega_3$$

Toffoli Gate (controlled-controlled-NOT gate): is a three-qubit gate which flips the right qubit if the left and middle qubits are 1.

$$\text{Toffoli}|a\rangle|b\rangle|c\rangle = |a\rangle|b\rangle|ab \oplus c\rangle.$$



Example 5.3.5: Construct the matrix representation of the Toffoli gate

$$\text{Toff}|\hat{0}00\rangle = |000\rangle$$

$$\text{Toff}|001\rangle = |001\rangle$$

$$\text{Toff}|010\rangle = |010\rangle$$

$$\text{Toff}|011\rangle = |011\rangle$$

$$\text{Toff}|100\rangle = |100\rangle$$

$$\text{Toff}|101\rangle = |101\rangle$$

$$\text{Toff}|110\rangle = |111\rangle$$

$$\text{Toff}|111\rangle = |110\rangle$$

$$\text{Toff}|abc\rangle = |ab, ab \oplus c\rangle$$

a	b	ab
0	0	0
0	1	0
1	0	0
1	1	1

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$